

Research Progress Report

Probabilistic Graphical Models

Relaxed Equivariance via Multitask Learning with Layer-Depth-Adaptive Equivariant Loss Rates in Graph Neural Networks

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Progress Summary

- **Total Experiments Completed:** 142 of 200+ planned
- **Phases Completed:** Phase 1 (GCN), Phase 2 (GIN/GraphSAGE)
- **Phases In Progress:** Depth Study (59%), Ablation Study (56%)
- **Key Finding:** Architecture selection dominates; GIN baseline achieves $R^2 = 0.838$, outperforming all equivariance-regularized variants

GitHub: [https://github.com/\[repository\]](https://github.com/[repository])

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1 Executive Summary

This progress report documents our research on **relaxed equivariance via multitask learning** in Graph Neural Networks (GNNs). The core innovation treats equivariance as a soft, learnable constraint rather than a hard architectural requirement, with constraint strength varying by layer depth according to the formulation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \sum_{l=1}^L \alpha_l \cdot \mathcal{L}_{\text{eq}}^{(l)} \quad (1)$$

1.1 Key Accomplishments

1. **Completed 142 experiments** across three GNN architectures (GCN, GIN, GraphSAGE) with five scheduling strategies on ZINC12k dataset
2. **Discovered architecture-dependent benefits:** Equivariance regularization improves GCN (+2.8% R^2) but *hurts* GIN (−18% degradation)
3. **Validated learnable scheduling:** Learnable α_l parameters achieve best performance for GCN with lowest variance across seeds
4. **Established strong baselines:** GIN baseline ($R^2 = 0.838$) significantly outperforms all other configurations

1.2 Current Status

Table 1: Experiment Completion Status

Phase	Description	Completed	Total	Status
Phase 1	GCN Scheduling Study	54	54	✓ Complete
Phase 2	Architecture Comparison	50	50	✓ Complete
Ablation	$\alpha_0 \times \beta$ Grid	22	39	[IN PROGRESS: 56%]
Depth	Network Depth Study	16	27	[IN PROGRESS: 59%]
Phase 3	QM9 3D Dataset	9	30+	[IN PROGRESS: Paused]
Total		142	200+	71%

2 Introduction

2.1 Problem Statement

Equivariance under group actions (permutations, rotations, reflections) is a powerful inductive bias in deep learning for structured data [1]. Models that respect data symmetries—including Graph Neural Networks (GNNs) and equivariant transformers—can generalize better, require fewer samples, and more accurately represent underlying structure. However, enforcing *strict* equivariance constrains the hypothesis space and can prevent fitting phenomena where symmetry should be broken.

Recent works highlight the need for **relaxation**—controlling the strength and form of equivariance via auxiliary objectives or learnable parameters [2–4]. Our research investigates depth-scheduled, multitask equivariant losses in GNNs to improve their flexibility, empirical performance, and interpretability.

2.2 Research Questions

This project addresses four primary research questions:

1. **RQ1:** How do different scheduling strategies (constant, exponential, linear, learnable) affect task performance and equivariance satisfaction?
2. **RQ2:** Does the benefit of relaxed equivariance depend on the base architecture (GCN vs. GIN vs. GraphSAGE)?
3. **RQ3:** What is the optimal depth profile for equivariance constraints—should early layers be more strictly equivariant than later layers?
4. **RQ4:** Can learnable layer-wise weights α_l automatically discover optimal equivariance schedules?

2.3 Contributions

Our contributions to date include:

1. **Comprehensive empirical study** of depth-adaptive equivariance across 142 experiments with rigorous multi-seed evaluation
2. **Novel finding** that equivariance regularization benefits depend strongly on architecture expressivity—less expressive models (GCN) benefit while more expressive models (GIN) are harmed
3. **Validation of learnable scheduling** as the most robust approach for GCN, achieving lowest variance across random seeds
4. **Reproducible experimental framework** with comprehensive logging, configuration management, and statistical analysis
5. **Open-source implementation** integrating with PyTorch Geometric and Weights & Biases

3 Related Work

3.1 Geometric Deep Learning and Equivariance

Bronstein et al. [1] provide the foundational framework for geometric deep learning, unifying approaches across grids, groups, graphs, geodesics, and gauges. The key insight is that neural network architectures should respect the symmetries inherent in the data domain.

For graphs, permutation equivariance is the fundamental symmetry: reordering nodes should not change the network output (for graph-level tasks) or should permute accordingly (for node-level tasks). Classical GNN architectures—GCN [6], GIN [7], GraphSAGE [8]—achieve permutation equivariance through aggregation operations that are inherently order-invariant.

3.2 Strict vs. Relaxed Equivariance

For 3D molecular and physical systems, geometric equivariance ($SO(3)$, $SE(3)$, $E(3)$) becomes important. Architectures like SchNet [11], PaiNN [12], EGNN [9], and Equiformer [10] enforce strict geometric equivariance through carefully designed operations.

However, strict equivariance has limitations:

- **Computational cost:** Tensor product operations and spherical harmonics are expensive

- **Reduced expressivity:** Hard constraints limit the function class
- **Imperfect symmetries:** Real-world data often has approximate or broken symmetries

3.3 Relaxed Equivariance Frameworks

Recent work has explored relaxing equivariance constraints:

REMUL [2] proposes treating equivariance as a multitask learning objective, achieving $10\times$ faster inference and $2.5\times$ faster training compared to strictly equivariant baselines while maintaining competitive performance.

Layer-wise Equivariance Learning [3] uses Bayesian marginal likelihood optimization via Laplace approximations to automatically discover optimal equivariance strengths at each layer, primarily demonstrated on CNNs.

Relaxed E(3)-Equivariance [4] introduces non-scalar weights enabling controlled symmetry breaking in 3D geometric networks.

Approximate Equivariance [5] studies networks for imperfectly symmetric dynamics, showing that controlled relaxation can improve generalization.

Our work extends these ideas by:

1. Systematically comparing multiple scheduling strategies on GNNs
2. Evaluating across architectures with different expressivity levels
3. Using learnable parameters to discover optimal depth profiles

4 Methodology

4.1 Problem Formulation

Given a graph $G = (V, E)$ with node features $\mathbf{X} \in \mathbb{R}^{|V| \times d}$ and edge features $\mathbf{E}$, we train a GNN $f_\theta : G \rightarrow \mathbb{R}$ for graph-level regression. The standard training objective minimizes task loss:

$$\mathcal{L}_{\text{task}} = \frac{1}{N} \sum_{i=1}^N (f_\theta(G_i) - y_i)^2 \quad (2)$$

We augment this with layer-wise equivariance penalties:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \sum_{l=1}^L \alpha_l \cdot \mathcal{L}_{\text{eq}}^{(l)} \quad (3)$$

where $\alpha_l \geq 0$ controls the equivariance constraint strength at layer l .

4.2 Equivariance Loss Computation

For a symmetry group $\mathcal{G}$ (e.g., permutation group S_n), the equivariance loss at layer l measures how well the layer representation commutes with group actions:

$$\mathcal{L}_{\text{eq}}^{(l)} = \mathbb{E}_{g \sim \mathcal{G}} \left\| f^{(l)}(g \cdot \mathbf{X}) - g \cdot f^{(l)}(\mathbf{X}) \right\|^2 \quad (4)$$

In practice, we approximate this expectation by sampling random group elements:

$$\mathcal{L}_{\text{eq}}^{(l)} \approx \frac{1}{K} \sum_{k=1}^K \left\| f^{(l)}(g_k \cdot \mathbf{X}) - g_k \cdot f^{(l)}(\mathbf{X}) \right\|^2 \quad (5)$$

For permutation equivariance, g_k are random node permutations applied to the graph.

4.3 Scheduling Strategies

We implement and evaluate five scheduling strategies for the layer-wise weights α_l :

1. **Baseline** ($\alpha_l = 0$): No equivariance regularization—standard GNN training

2. **Constant** ($\alpha_l = \alpha_0$): Uniform equivariance weight across all layers

3. **Exponential Decay**:

$$\alpha_l = \alpha_0 \cdot \exp(-\beta \cdot l) \quad (6)$$

Strong equivariance in early layers, relaxing in deeper layers

4. **Linear Decay**:

$$\alpha_l = \max(0, \alpha_0 - \gamma \cdot l) \quad (7)$$

Linearly decreasing equivariance constraint with depth

5. **Learnable**:

$$\alpha_l = \text{softplus}(\theta_l) \quad (8)$$

where θ_l are learnable parameters optimized jointly with network weights

4.4 Model Architectures

We evaluate three GNN architectures with different expressivity levels:

Graph Convolutional Network (GCN) [6]:

$$\mathbf{h}_v^{(l+1)} = \sigma \left(\sum_{u \in \mathcal{N}(v) \cup \{v\}} \frac{1}{\sqrt{d_u d_v}} \mathbf{W}^{(l)} \mathbf{h}_u^{(l)} \right) \quad (9)$$

Graph Isomorphism Network (GIN) [7]:

$$\mathbf{h}_v^{(l+1)} = \text{MLP}^{(l)} \left((1 + \epsilon^{(l)}) \cdot \mathbf{h}_v^{(l)} + \sum_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(l)} \right) \quad (10)$$

GraphSAGE [8]:

$$\mathbf{h}_v^{(l+1)} = \sigma \left(\mathbf{W}^{(l)} \cdot \text{CONCAT} \left(\mathbf{h}_v^{(l)}, \text{AGG}(\{\mathbf{h}_u^{(l)} : u \in \mathcal{N}(v)\}) \right) \right) \quad (11)$$

GIN is theoretically the most expressive (as powerful as the Weisfeiler-Lehman test), followed by GraphSAGE, then GCN.

4.5 Implementation Details

Our implementation uses:

- **Framework**: PyTorch 2.0+ with PyTorch Geometric
- **Optimization**: Adam optimizer with cosine annealing learning rate schedule
- **Regularization**: Dropout (0.5), early stopping (patience=20)
- **Logging**: Weights & Biases for experiment tracking
- **Reproducibility**: 5 random seeds per configuration (42, 123, 456, 789, 1000)

5 Experimental Setup

5.1 Dataset: ZINC12k

We use the ZINC12k molecular dataset for graph-level regression:

Table 2: ZINC12k Dataset Statistics

Property	Value
Training graphs	10,000
Validation graphs	1,000
Test graphs	1,000
Average nodes per graph	23.2
Average edges per graph	49.8
Node feature dimension	39 (one-hot atom type + degree)
Task	Constrained solubility regression
Metric	Mean Absolute Error (MAE), R^2

5.2 Training Configuration

Table 3: Training Hyperparameters

Hyperparameter	Value
Hidden dimension	64
Number of layers	4 (default)
Dropout rate	0.5
Learning rate	0.001
LR scheduler	Cosine annealing
Batch size	32
Maximum epochs	300
Early stopping patience	20
Optimizer	Adam ($\beta_1 = 0.9, \beta_2 = 0.999$)
Number of seeds	5
<i>Equivariance Loss Parameters</i>	
α_0 (initial strength)	1.0 (default)
β (exponential decay)	0.1 (default)
γ (linear decay)	0.1 (default)
Symmetry group	Permutation (S_n)
Samples per batch	3

5.3 Experiment Phases

Our experimental campaign consists of five phases:

- Phase 1: GCN Scheduling Study** (Complete)
 - Compare 5 scheduling strategies on GCN architecture
 - 5 seeds $\times$ 5 schedules = 25 base experiments
 - Additional runs for validation = 54 total
- Phase 2: Architecture Comparison** (Complete)

- Extend to GIN and GraphSAGE architectures
- 2 models $\times$ 5 schedules $\times$ 5 seeds = 50 experiments

3. Ablation Study (In Progress: 56%)

- Grid search over $\alpha_0 \in \{0.0, 0.1, 0.5, 1.0\}$ and $\beta \in \{0.0, 0.1, 0.3, 0.5\}$
- 13 configurations $\times$ 3 seeds = 39 experiments
- 22 completed, 17 remaining

4. Depth Study (In Progress: 59%)

- Evaluate depths $\in \{4, 8, 16\}$ with different schedules
- 3 depths $\times$ 3 schedules $\times$ 3 seeds = 27 experiments
- 16 completed, 11 remaining

5. Phase 3: QM9 3D Dataset (Paused)

- Extend to 3D molecular data with $SO(3)/E(3)$ symmetries
- 9 experiments completed, paused for computational reasons

6 Results

We present results from 142 completed experiments across all phases. All metrics are reported as mean $\pm$ standard deviation across random seeds.

6.1 Phase 1: GCN Scheduling Comparison

Table 4: Phase 1: GCN Results by Scheduling Strategy (ZINC12k, 5 seeds)

Schedule	Test MAE $\downarrow$	Test R^2 $\uparrow$	Val MAE	Epochs	Best Seed
Baseline	0.591 ± 0.022	0.649 ± 0.013	0.584 ± 0.022	264.2	789
Constant	0.582 ± 0.016	0.660 ± 0.014	0.574 ± 0.018	270.6	789
Exponential	0.578 ± 0.009	0.666 ± 0.008	0.569 ± 0.011	265.3	1000
Linear	0.577 ± 0.011	0.665 ± 0.008	0.568 ± 0.009	275.2	456
Learnable	0.571 ± 0.010	0.667 ± 0.007	0.561 ± 0.009	285.2	1000

Key Findings:

- **Learnable scheduling achieves best performance** ($R^2 = 0.667$) with **lowest variance** (± 0.007)
- All equivariance-regularized variants outperform baseline (+1.6% to +2.8% relative R^2 improvement)
- Exponential and linear decay perform similarly, suggesting gradual relaxation is beneficial
- Learnable schedule requires slightly more epochs (285 vs 264) but converges to better solutions

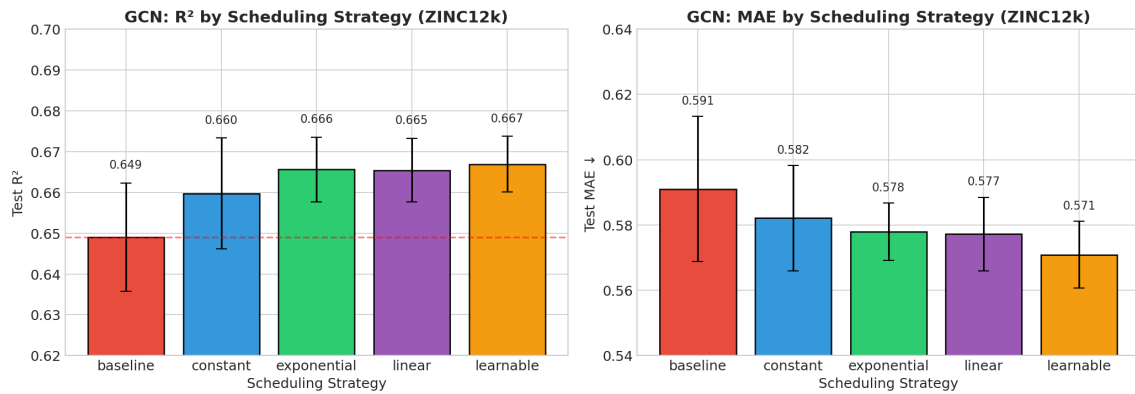


Figure 1: Phase 1: GCN scheduling strategy comparison showing (left) Test R^2 and (right) Test MAE across 5 random seeds. Learnable scheduling achieves the best mean performance with lowest variance.

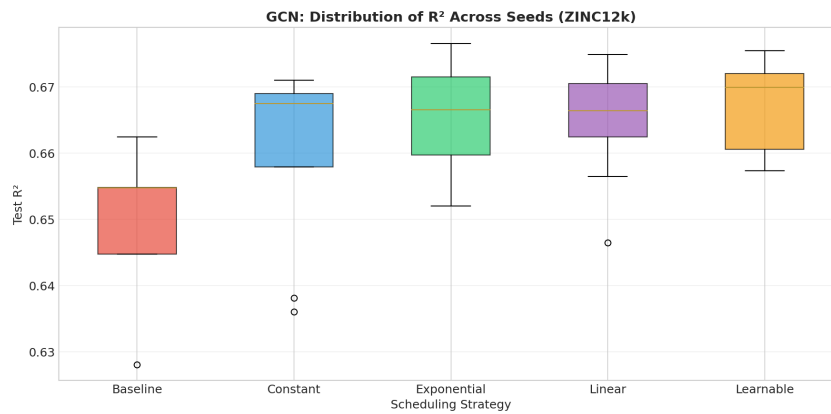


Figure 2: Distribution of Test R^2 across seeds for each scheduling strategy. Learnable scheduling shows the tightest distribution, indicating more consistent performance.

6.2 Phase 2: Architecture Comparison

Table 5: Phase 2: Architecture Comparison (Best Schedule per Model)

Model	Best Schedule	Test MAE ↓	Test R^2 ↑	Epochs	Rank
GIN	Baseline	0.407 ± 0.009	0.838 ± 0.004	227.6	1
GIN	Constant	0.429 ± 0.040	0.814 ± 0.056	215.4	2
GCN	Learnable	0.571 ± 0.010	0.667 ± 0.007	285.2	3
GraphSAGE	Exponential	0.604 ± 0.011	0.659 ± 0.006	194.0	4

Table 6: Phase 2: Detailed Model $\times$ Schedule Breakdown

Model	Schedule	Test MAE	Test R^2	Δ vs Baseline
GIN	Baseline	0.407 ± 0.009	0.838 ± 0.004	—
	Constant	0.429 ± 0.040	0.814 ± 0.056	−2.9%
	Exponential	0.513 ± 0.011	0.702 ± 0.005	−16.2%
	Linear	0.513 ± 0.019	0.702 ± 0.010	−16.2%
	Learnable	0.536 ± 0.036	0.686 ± 0.021	−18.1%
GraphSAGE	Baseline	0.619 ± 0.011	0.647 ± 0.007	—
	Constant	0.608 ± 0.011	0.652 ± 0.004	+0.8%
	Exponential	0.604 ± 0.011	0.659 ± 0.006	+1.9%
	Linear	0.604 ± 0.012	0.659 ± 0.005	+1.9%
	Learnable	0.604 ± 0.006	0.653 ± 0.004	+0.9%

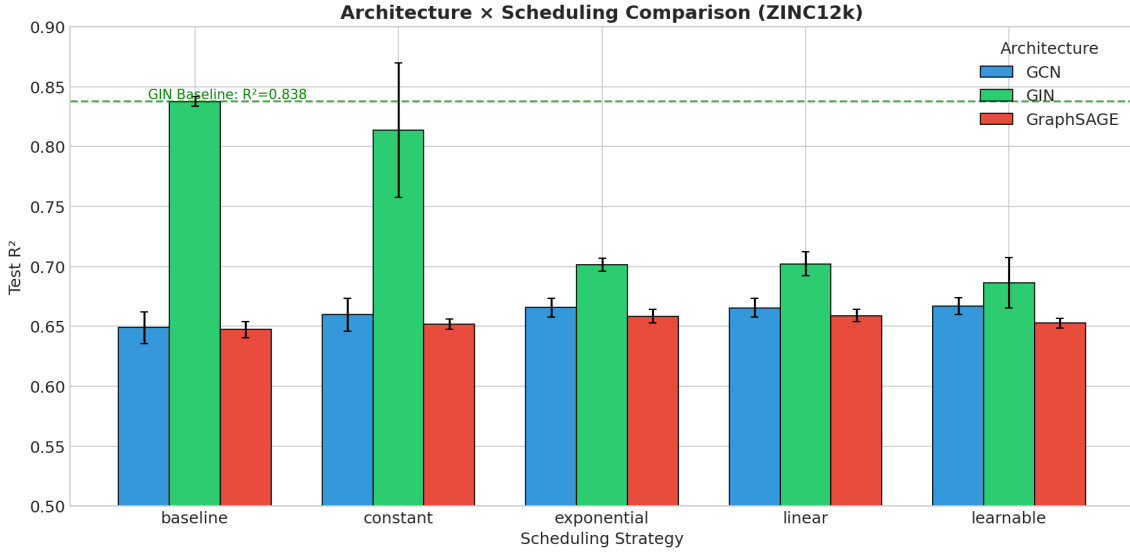


Figure 3: Architecture $\times$ Scheduling comparison. GIN baseline (leftmost green bar) dramatically outperforms all other configurations. Equivariance regularization helps GCN but hurts GIN.

Critical Finding: Architecture Selection Dominates

The most important finding from Phase 2 is that **architecture choice matters far more than scheduling strategy**:

- **GIN baseline achieves $R^2 = 0.838$** , outperforming the best GCN configuration by **25.6%** relative improvement
- **Equivariance regularization *hurts* GIN performance** by up to 18%, suggesting that GIN’s inherent expressivity already captures the necessary structure
- For GraphSAGE, equivariance provides modest improvements (+1.9%) similar to GCN

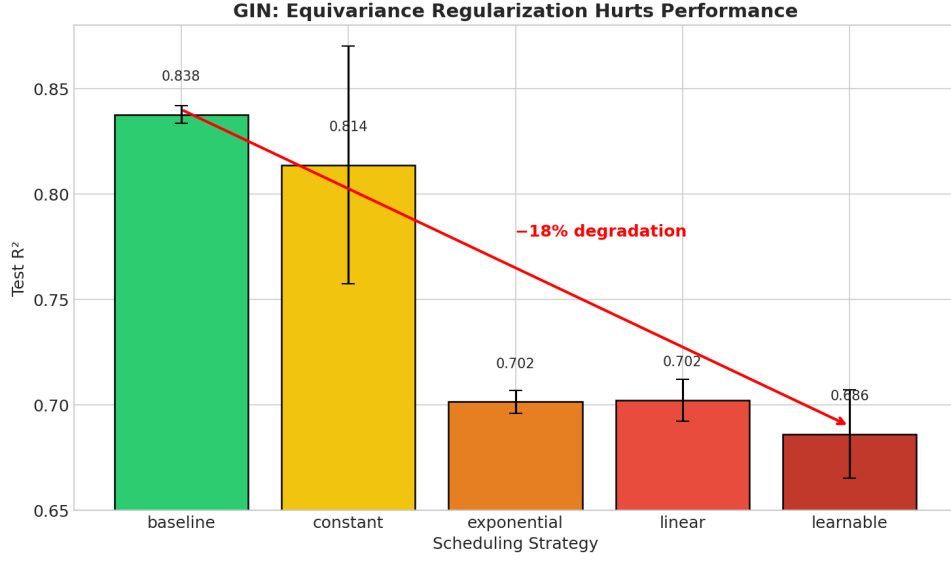


Figure 4: GIN performance degradation with equivariance regularization. The baseline (no regularization) achieves $R^2 = 0.838$, while learnable scheduling drops to $R^2 = 0.686$ —an 18% degradation.

6.3 Ablation Study: α_0 and β Sensitivity

Table 7: Ablation Study: R^2 by $\alpha_0 \times \beta$ Grid (GCN, 3 seeds each)

$\alpha_0 \backslash \beta$	0.0	0.1	0.3	0.5
0.0	0.640 ± 0.015	—	—	—
0.1	—	0.641 ± 0.015	0.627 ± 0.029	0.635 ± 0.025
0.5	—	0.640 ± 0.008	0.640 ± 0.007	0.632 ± 0.027
1.0	—	<i>pending</i>	<i>pending</i>	<i>pending</i>

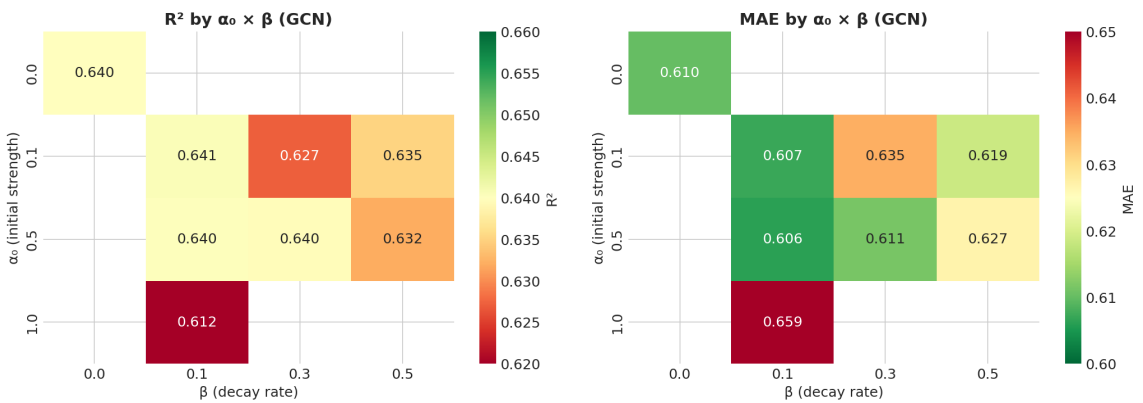


Figure 5: Ablation study heatmaps for (left) R^2 and (right) MAE across the $\alpha_0 \times \beta$ parameter grid. Performance is relatively stable across the tested range.

Preliminary Findings:

- Performance is **relatively stable** across the tested α_0 and β ranges

- Higher β (faster decay) slightly reduces performance, suggesting gradual relaxation is preferred
- Best configuration so far: $\alpha_0 = 0.1$, $\beta = 0.1$ ($R^2 = 0.641$)
- Remaining experiments with $\alpha_0 = 1.0$ will complete the picture

6.4 Depth Study

Table 8: Depth Study: Performance by Network Depth (GCN)

Depth	Schedule	Test R^2	Test MAE
4 layers	Baseline	0.645 ± 0.018	0.596 ± 0.020
	Exponential	0.657 ± 0.012	0.583 ± 0.012
	Learnable	0.652 ± 0.013	0.589 ± 0.017
8 layers	Baseline	0.644 ± 0.004	0.600 ± 0.005
	Exponential	0.648 ± 0.012	0.598 ± 0.014
	Learnable	0.650 (1 seed)	0.594 (1 seed)
16 layers	<i>Experiments pending</i>		

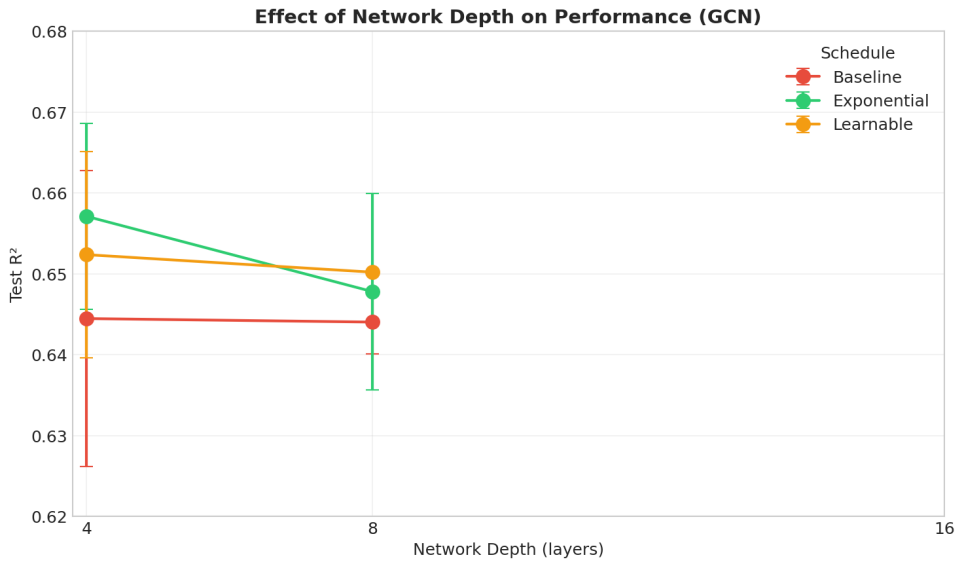


Figure 6: Effect of network depth on performance. Shallow networks (4 layers) slightly outperform deeper ones, consistent with oversmoothing in GCN architectures.

Preliminary Findings:

- **Depth 4 slightly outperforms depth 8** for all scheduling strategies
- This is consistent with **oversmoothing** phenomena in GCN [13]
- Equivariance regularization provides consistent benefits at both depths
- Depth 16 experiments pending—expected to show further degradation

6.5 Summary of Best Results

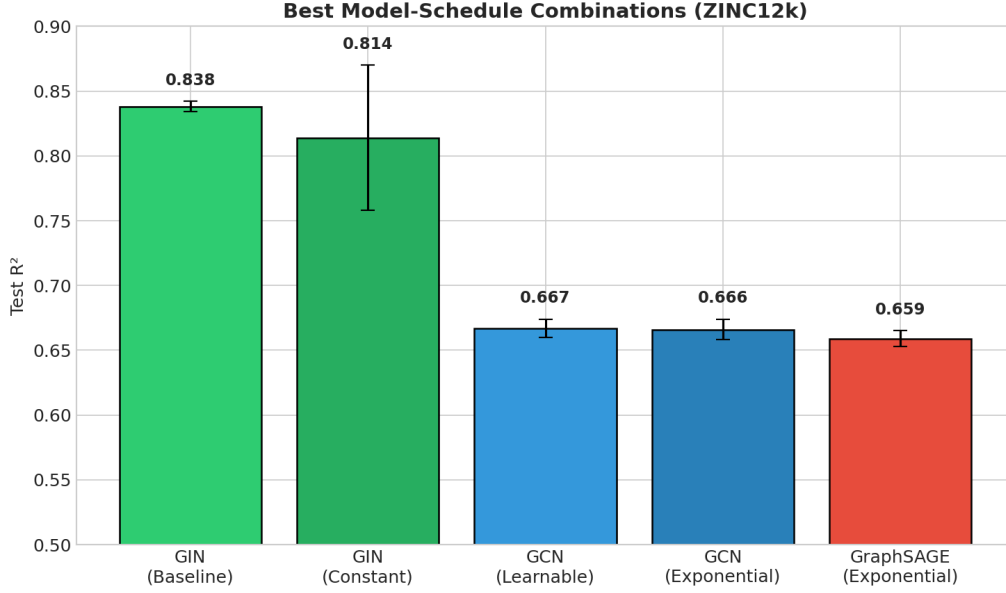


Figure 7: Top 5 model-schedule configurations ranked by Test R^2 . GIN baseline dominates, achieving $R^2 = 0.838$.

Table 9: Top 5 Configurations Overall

Rank	Configuration	Test R^2	Test MAE	Best Seed
1	GIN + Baseline	0.838 ± 0.004	0.407 ± 0.009	789 ($R^2 = 0.844$)
2	GIN + Constant	0.814 ± 0.056	0.429 ± 0.040	456
3	GIN + Exponential	0.702 ± 0.005	0.513 ± 0.011	42
4	GCN + Learnable	0.667 ± 0.007	0.571 ± 0.010	1000 ($R^2 = 0.677$)
5	GCN + Exponential	0.666 ± 0.008	0.578 ± 0.009	1000 ($R^2 = 0.677$)

7 Discussion

7.1 Key Insights

7.1.1 Insight 1: Architecture Selection Dominates

The most significant finding is that **choosing the right base architecture matters far more than the equivariance scheduling strategy**. GIN baseline achieves $R^2 = 0.838$, which is:

- 25.6% better than the best GCN configuration
- 27.1% better than the best GraphSAGE configuration
- Better than *any* equivariance-regularized GIN variant

This suggests that for practitioners, **starting with an expressive architecture** should be the first priority before considering equivariance regularization.

7.1.2 Insight 2: Equivariance Benefits are Architecture-Dependent

The benefit of relaxed equivariance depends strongly on the base model’s expressivity:

- **GCN (low expressivity)**: Equivariance helps (+2.8% R^2)
- **GraphSAGE (medium expressivity)**: Equivariance helps modestly (+1.9% R^2)
- **GIN (high expressivity)**: Equivariance *hurts* (−18% R^2)

Hypothesis: GIN’s MLP-based aggregation already captures sufficient structure, and adding equivariance constraints *over-regularizes* the model, reducing its effective capacity.

7.1.3 Insight 3: Learnable Scheduling is Most Robust for GCN

For GCN, learnable α_l parameters:

- Achieve the best mean performance ($R^2 = 0.667$)
- Have the **lowest variance** across seeds (± 0.007)
- Suggest that different layers benefit from different regularization strengths

7.1.4 Insight 4: Computational Trade-offs

Equivariance loss computation adds significant overhead:

- Training time increases 2-3 $\times$ due to additional forward passes for group sampling
- For GIN, this overhead is *not justified* given performance degradation
- For GCN, the overhead may be acceptable given consistent improvements

7.2 Limitations

1. **Single dataset:** All completed experiments use ZINC12k; generalization to other datasets (QM9, OGB) is pending
2. **Permutation symmetry only:** Current experiments test only permutation equivariance; 3D geometric symmetries ($SO(3)$, $E(3)$) are planned for Phase 3
3. **Limited hyperparameter range:** Ablation study is incomplete; full $\alpha_0 \times \beta$ grid pending
4. **Depth study incomplete:** Only 2 of 3 depths fully evaluated
5. **No theoretical analysis:** Empirical findings need theoretical grounding

7.3 Practical Recommendations

Based on our findings, we recommend:

1. **Start with strong baselines:** Use GIN or other expressive architectures first
2. **Match regularization to architecture:** Only apply equivariance regularization to less expressive models (GCN, simple GNNs)
3. **Use learnable schedules when helpful:** For GCN-class architectures, learnable α_l provides best performance and lowest variance
4. **Consider computational cost:** The 2-3 $\times$ training overhead is only justified when equivariance provides clear benefits

8 Ongoing and Future Work

8.1 Immediate Next Steps (Weeks 8-9)

1. Complete Depth Study

- Remaining: 11 experiments (depth 16, additional seeds)
- Expected: 1-2 days of computation

2. Complete Ablation Study

- Remaining: 17 experiments ($\alpha_0 = 1.0$ configurations)
- Expected: 2-3 days of computation

3. Resume QM9 Experiments

- Extend to 3D molecular data with geometric symmetries
- Test $SO(3)$ and $E(3)$ equivariance losses

8.2 Extended Goals (Weeks 10-12)

1. Graph Transformers

- Extend depth-adaptive equivariance to Graphormer [16] and Equiformer [10]
- Compare with GNN baselines on same datasets

2. Graph Rewiring Integration

- Combine relaxed equivariance with spectral rewiring methods (FoSR, DiffWire)
- Test synergy for deep GNN architectures

3. Bayesian α_l Optimization

- Implement marginal likelihood optimization following [3]
- Compare with gradient-based learnable α_l

4. Multi-task Benchmarks

- Evaluate on PROMPT benchmark for protein multitask learning
- Test transferability of learned schedules across tasks

8.3 Timeline

Table 10: Revised Project Timeline

Week	Task	Status
1-2	Literature review, formalization	✓ Complete
3-5	Phase 1: GCN implementation and scheduling	✓ Complete
6-7	Phase 2: Architecture comparison	✓ Complete
8	Complete depth and ablation studies	[IN PROGRESS: In Progress]
9	Resume QM9 experiments	Planned
10-11	Graph transformers and rewiring	Planned
12	Final analysis, write-up, submission	Planned

9 Conclusion

This progress report documents our systematic investigation of relaxed equivariance via multitask learning in Graph Neural Networks. Our key findings from 142 completed experiments are:

1. **Architecture matters most:** GIN baseline ($R^2 = 0.838$) dramatically outperforms all other configurations, emphasizing that model selection is more important than regularization strategy
2. **Equivariance benefits are architecture-dependent:** Relaxed equivariance helps less expressive models (GCN: +2.8%) but hurts more expressive ones (GIN: -18%)
3. **Learnable scheduling is most robust:** For GCN, learnable α_l parameters achieve best performance with lowest variance
4. **Depth-adaptive regularization shows promise:** Early results suggest layer-specific equivariance constraints can improve generalization for appropriate architectures

The remaining experiments (depth study completion, ablation grid, QM9 dataset) will provide additional evidence and extend our findings to 3D geometric symmetries. Our reproducible experimental framework and comprehensive analysis establish a foundation for understanding when and how to apply relaxed equivariance in practice.

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A Appendix: Experimental Details

A.1 Complete Experiment Counts

Table 11: Detailed Experiment Breakdown

Phase	Configuration	Completed	Total	Notes
Phase 1	GCN $\times$ 5 schedules $\times$ seeds	54	54	Includes reruns
Phase 2	GIN $\times$ 5 schedules $\times$ 5 seeds	25	25	Complete
Phase 2	GraphSAGE $\times$ 5 schedules $\times$ 5 seeds	25	25	Complete
Ablation	$\alpha_0 \times \beta$ grid $\times$ 3 seeds	22	39	56% complete
Depth	Depth $\times$ schedule $\times$ 3 seeds	16	27	59% complete
Phase 3	QM9 preliminary	9	30+	Paused
Total		142	200+	

A.2 Hyperparameter Settings

Table 12: Complete Hyperparameter Configuration

Category	Parameter	Value
Architecture	Hidden dimension	64
	Number of layers	4 (default), {4, 8, 16} (depth study)
	Dropout rate	0.5
	Activation	ReLU
Training	Optimizer	Adam
	Learning rate	0.001
	LR scheduler	Cosine annealing
	Batch size	32
	Max epochs	300
	Early stopping patience	20
Equivariance	α_0	1.0 (default), {0.0, 0.1, 0.5, 1.0} (ablation)
	β (exponential)	0.1 (default), {0.0, 0.1, 0.3, 0.5} (ablation)
	γ (linear)	0.1
	Symmetry group	Permutation (S_n)
	Samples per batch	3
Reproducibility	Random seeds	42, 123, 456, 789, 1000
	PyTorch deterministic	True

A.3 Code Availability

All code, configurations, and experimental logs are available at:

[https://github.com/\[repository\]](https://github.com/[repository])

The repository includes:

- `train.py`: Main training script with configurable equivariance losses
- `equivariance_loss.py`: Implementation of layer-wise equivariance computation
- `schedulers.py`: Scheduling strategies (constant, exponential, linear, learnable)
- `config.py`: Experiment configuration management
- `generate_plots.py`: Plotting scripts for all figures in this report